University of Glasgow

# ENG4200 Intro to AI/ML CW1 Report

Dataset: Stroke Prediction

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## 1 Objective

This exercise applies logistic regression to a stroke classification task: predicting whether a patient has had a stroke from information about demographic, lifestyle and clinical information. The stroke prediction dataset contains 5,110 patient records with 10 features, and only 249 (4.87%) are stroke cases, so a model that always predicts 'no stroke' already scores 95.13% accuracy. The aim was to test whether class weighting and regularisation could turn this into a model that actually finds stroke patients, judged on recall, precision, F1 and ROC-AUC rather than accuracy, and to measure its computational cost in fit time, prediction time and memory.

## 2 Pipeline Design

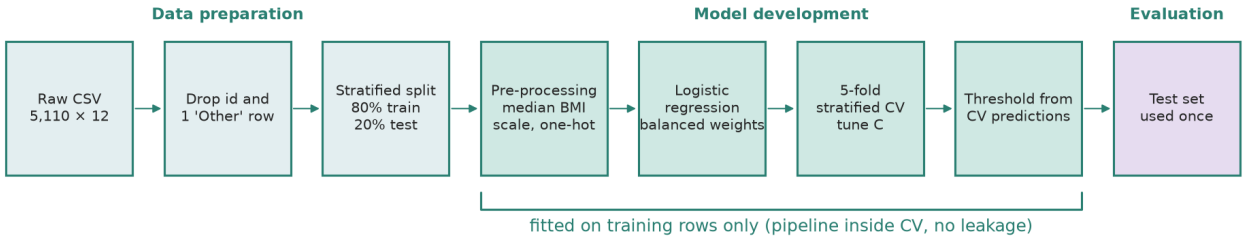



Figure 1: Pipeline from raw data to the final evaluation. Every step under the bracket only sees training rows.

**Data investigation.** BMI is missing for 201 patients (3.9%), but 40 of them had a stroke, which is 16.06% of all stroke cases, so dropping these rows would remove too much of the minority class. The missing values were instead filled with the median BMI rather than the mean (28.10 and 28.89 on the full data; the model learns the median from the training rows only), because BMI is positively skewed (skew 1.06, with values up to 97.6; Figure 8). The ID column was dropped, as was the single 'Other' gender row, which had no stroke cases and so carries no information for stroke detection. 'Unknown' smoking status (1,544 patients, 30%) was kept as its own category: it is not random, as these patients are much younger (mean age 30) and many are children, so filling it in would invent smoking histories for children. Range checks found no impossible values: the binary columns contain only 0 and 1, and the 115 non-integer ages (at most 1.88) are infants recorded in months, so they are not errors. Stroke

rate rises sharply with age, from 0.1% in the 18–33 band to 17.4% in the 68–89 band (UKHSA age bands), and stroke patients also show a second peak at high glucose levels (Figure 2).

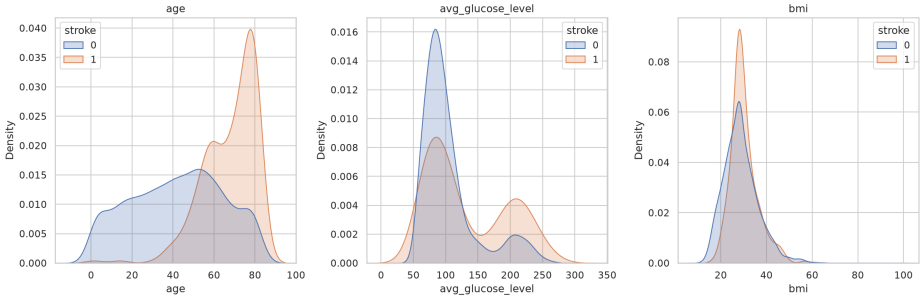



Figure 2: Distributions of age, average glucose level and BMI for patients without (0) and with (1) a stroke.

Stroke rates are also much higher with hypertension (about 13% vs 4%) and heart disease (about 17% vs 4%), while ever-married patients (6.6% vs 1.7%) and children (almost 0%) mostly reflect age (Figure 7). No numeric feature correlates strongly with stroke (age is highest at 0.25, BMI lowest at 0.04), and no pair of features is strongly correlated (maximum 0.32, age and BMI; Figure 9), so collinearity is not a concern for logistic regression and all features were kept. All ages were also kept to avoid biasing the model. Most exploration figures are in the Appendix.

**Model pipeline.** Figure 1 shows the full pipeline. The data was split 80/20 with stratification (`random_state=8`), giving 4,087 training and 1,022 test patients with the same stroke share (4.87% and 4.89%). Inside the sklearn Pipeline, numeric features are median-imputed and standardised, and categorical features are one-hot encoded, giving 20 features. Because the pipeline is refitted inside every cross-validation fold, the median and scaling are learned from training rows only, which prevents data leakage. Logistic regression with `class_weight='balanced'` was tuned with 5-fold stratified cross-validation over seven values of $C$ from $10^{-3}$ to $10^{3}$, scored on F1. The decision threshold was chosen from cross-validated probabilities on the training rows, and no decisions were made on the test set, which was used only for the final evaluation.

## 3 Key Results

Table 1 compares the experiments with 5-fold cross-validation on the training rows, followed by the single test-set evaluation. Accuracy is shown but not used to choose the model, because the unweighted model matches the 95.13% majority baseline while finding almost no strokes. All metrics come from the classification confusion matrix: recall is the share of real strokes the model catches, precision is the share of flagged patients who really had a stroke, and F1 is their harmonic mean. F1 runs from 0 to 1 (best) and is only high when both are high. ROC-AUC is threshold-free: it measures how well the predicted probabilities rank stroke patients above non-stroke patients.

Class weighting made the biggest difference: cross-validated recall rose from 0.005 to 0.789 and F1 from 0.010 to 0.228, while tuning $C$ added only a little (F1 0.234).

**Threshold.** The decision threshold was chosen from cross-validated probabilities on the training rows (Figure 3). Lowering the threshold raises recall but flags more patients for a follow-up. A threshold of 0.5 flags the fewest patients (29.5%) while keeping recall above 0.8 (0.824); raising it to 0.6 would cut recall to 0.683 with only a small gain in precision (0.136 to 0.160).

**Test set.** On the test set the unweighted model missed all 50 strokes, whereas the final model caught 36 of them (Figure 4), with a ROC-AUC of 0.826. The cost is 235 false alarms, so about 1 in 7.5

| Model | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| *5-fold cross-validation, training rows (4,087 patients)* | | | | |
| Majority baseline (always 'no stroke') | 0.951 | – | 0.000 | 0.000 |
| Baseline LogReg, unweighted ($C = 1$) | 0.951 | 0.200 | 0.005 | 0.010 |
| LogReg, balanced ($C = 1$) | 0.739 | 0.133 | 0.789 | 0.228 |
| LogReg, balanced, tuned ($C = 0.01$) | 0.737 | 0.137 | 0.824 | 0.234 |
| *Held-out test set (1,022 patients, 50 strokes), threshold 0.5* | | | | |
| Baseline LogReg, unweighted ($C = 1$) | 0.951 | 0.000 | 0.000 | 0.000 |
| Final model (balanced, $C = 0.01$) | 0.76 | 0.13 | 0.72 | 0.22 |

Table 1: Performance of each experiment. LogReg - logistic regression classifier. The majority baseline never predicts a stroke, so its precision is undefined (0/0) and its recall and F1 are zero by definition.

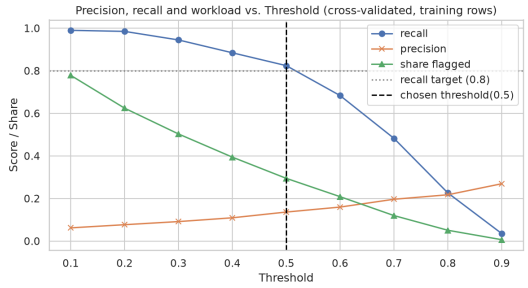



Figure 3: Cross-validated precision, recall and share flagged against the decision threshold (training rows).

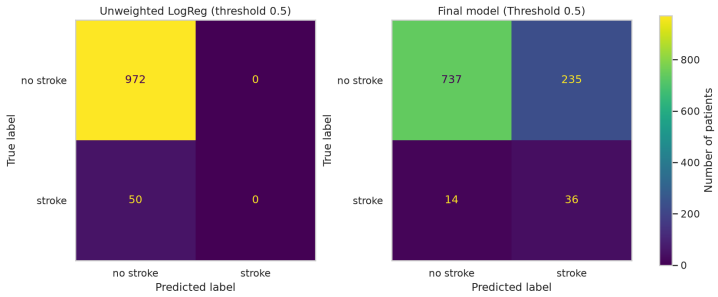



Figure 4: Test-set confusion matrices for the unweighted model and the final balanced model.

flagged patients had a stroke; in screening, a missed stroke is far worse than an extra check, so this trade-off is acceptable.

**Efficiency.** The hardware used for this exercise is a Lenovo Legion ACH6H (AMD Ryzen 7 5800H) with 32GB RAM, running the Linux distro Pop!_OS with Python 3.12.3 and sklearn 1.9.1; timings vary between runs and with background load, so they are approximate. Fitting the final model on 4,087 rows took about 19–20 ms, predicting 1,022 patients took about 6 ms, and peak Python memory during fitting (measured with `tracemalloc`, which does not count all C-level allocations) was 1.61 MB. Balanced weighting made no meaningful difference to cost (unweighted: 22.4 ms fit, 5.6 ms predict, 1.60 MB). Fit time grew roughly linearly with training size, from about 13 ms at 650 rows to 18.5 ms at 3,270 rows (Figure 6), but this is only 1.4 times the cost for 5 times the rows, so a fixed overhead of about 11 ms dominates and the scaling evidence is limited.

# 4 Analysis & Reflection

## What worked well

Balanced class weights were the single most effective change, turning a model that never predicted a stroke into one with a cross-validated recall of 0.824 at no extra computational cost. Keeping preprocessing inside the pipeline made every cross-validation score leakage-free. The coefficients (Figure 5) are easy to interpret and agree with the exploration (Figure 7): age is by far the strongest predictor, followed by average glucose level, hypertension and heart disease.

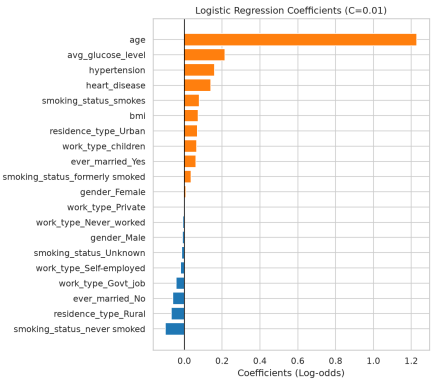



Figure 5: Coefficients of the final model on standardised features. Positive values raise the predicted stroke risk.

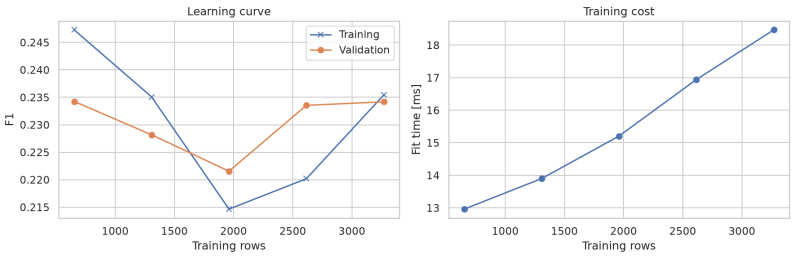



Figure 6: Learning curve (left) and fit time (right) against the number of training rows, with 5-fold cross-validation.

### Difficulties and limitations

The class imbalance was the main difficulty: even after weighting, precision stays low (0.13), so most flagged patients did not have a stroke. Balanced weighting also shifts the predicted probabilities upwards, so they were used only to choose a threshold, not as a patient's real chance of a stroke. The test set contains only 50 strokes, so each one changes recall by 0.02; this helps explain why test recall (0.72) is lower than cross-validated recall (0.824). Finally, logistic regression can only draw a linear decision boundary, and the flat learning curve suggests that this, together with the available features, limits performance more than the amount of data.

### Overfitting

Overfitting was addressed with stratified 5-fold cross-validation, preprocessing fitted on training rows only, a test set held back until the end, and $L_2$ regularisation tuned through $C$. Cross-validated F1 stayed between 0.228 and 0.234 for every $C$ from 0.001 to 1000, differences smaller than the spread between folds (std $\approx 0.01$), and training and validation F1 were almost identical (0.235 vs 0.234 at $C = 0.01$). The learning curve (Figure 6) confirms this: training and validation F1 converge at about 0.235 at full size. The model is therefore not overfitting, and $C = 0.01$ was chosen as the simpler of several effectively tied models.

## 5 Next Steps

With more time, a non-linear model such as the random forest, gradient boosting, or interaction terms such as age $\times$ glucose, would be tested to address the linear decision boundary. An L1 regularisation path would show which weak features could be removed, and the predicted probabilities could be calibrated so they can be read as a real stroke risk. Since the learning curve showed more rows alone are unlikely to help, new clinical features and an external stroke dataset could be used to improve the model and check whether it generalises.

# Appendix

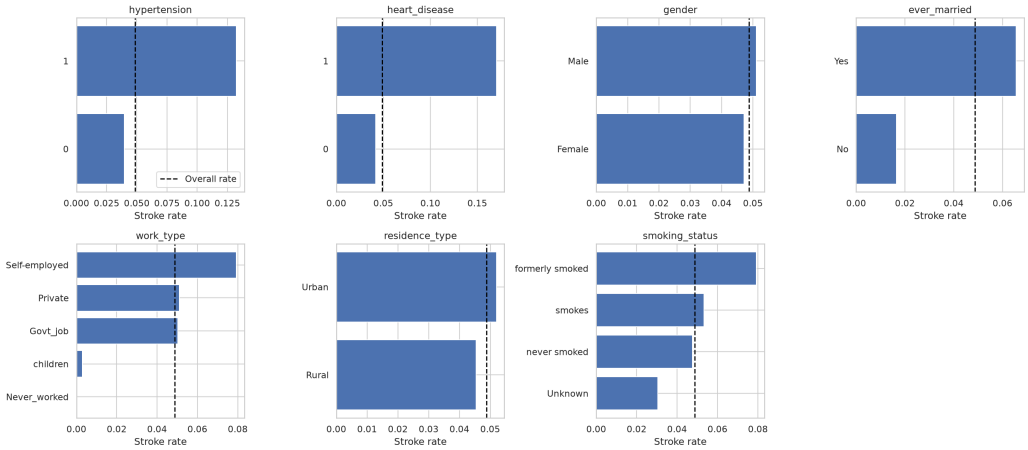



Figure 7: Stroke rate within each category. The dashed line is the overall stroke rate (4.87%).

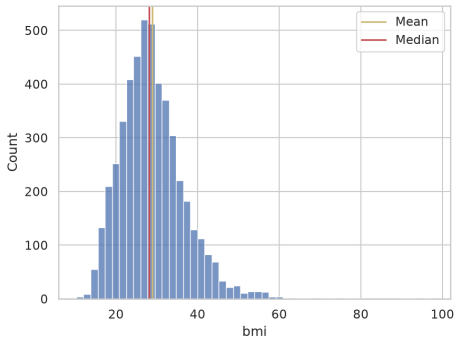



Figure 8: BMI distribution with the mean and median. The long right tail pulls the mean above the median.

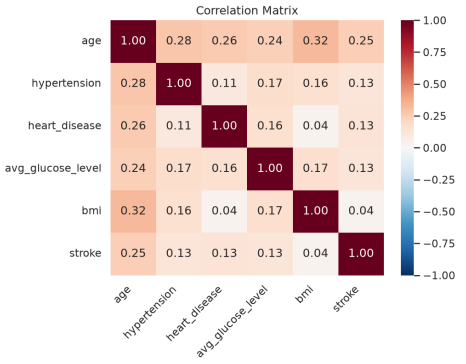



Figure 9: Correlation between the numeric features and stroke.